

Boomika S

MCA Student | AI & Full-Stack Developer

[Email](#) | [GitHub](#) | [LinkedIn](#) | [Portfolio](#) | +919345006355

SUMMARY

MCA student at Coimbatore Institute of Technology with hands-on experience building data-driven and AI-integrated applications. Interested in applying Python, machine learning, and data analysis to real-world business problems.

TECHNICAL SKILLS

Languages: Java, JavaScript, Python

Data & ML: Pandas, scikit-learn, NumPy, Machine Learning, Generative AI, Google Gemini API

Databases: SQL, DBMS, MongoDB, PostgreSQL

CS Fundamentals: Data Structures, Algorithms, OOP

Backend: REST APIs, Flask, Prisma

Web: HTML, CSS, Vanilla JS, React, Next.js

Tools: Git, GitHub, VS Code, Postman

PROJECTS

Personal Carbon Footprint Predictor [GitHub Repo](#)

Python | Flask | scikit-learn | Pandas | Gemini API

- Built a Flask web app predicting carbon footprint (kg CO₂e) using Linear Regression and Random Forest models trained through scikit-learn pipelines with a reusable preprocessing (ColumnTransformer) layer
- Trained a Random Forest classifier to categorize impact level (Low/Medium/High) alongside the regression models, evaluating performance with R², MAE, and accuracy metrics
- Integrated Google Gemini's API to generate personalized, context-aware reduction recommendations from a user's transport, energy, and diet data, with automatic fallback to rule-based logic if the API is unavailable
- Implemented user authentication and persistent JSON-based storage so footprint history is saved across sessions

AI Media SaaS Platform [GitHub Repo](#)

Next.js 14 | Prisma | PostgreSQL (NeonDB) | Cloudinary | Clerk

- Built a full-stack media management SaaS with Clerk authentication (email + Google) and a Prisma-modeled PostgreSQL database on NeonDB
- Integrated Cloudinary for video upload and automatic compression (up to 70MB), plus one-click image transformation into platform-specific formats (Instagram Square/Portrait, Twitter Post/Header)
- Designed the App Router architecture end-to-end — upload, preview-on-hover browsing, download, and delete flows

EDUCATION

Coimbatore Institute of Technology (CIT), Coimbatore

Aug 2025 – May 2027

Master of Computer Applications (MCA) | GPA: 8.45 (Sem I) | GPA: 9.00 (Sem II)

Sri Ramakrishna College of Arts & Science (SNR), Coimbatore

Aug 2022 – May 2025

B.Com – Computer Applications | CGPA: 8.89

ABC Matriculation Higher Secondary School, Coimbatore

HSC | 93%

Apr 2022

ABC Matriculation Higher Secondary School, Coimbatore

SSLC | 77%

Apr 2020

CERTIFICATIONS & ACHIEVEMENTS

- Full Stack Python - Edukators (hands-on, project-based)
- Product and Brand Management - NPTEL / SWAYAM (Govt. of India)
- Placement Coordinator, B.Com (CA), SNR – Coordinated between students, faculty members and placement drives
- Rank Holder - 2nd in department, B.Com (CA), SNR (2022–2025)
- Best Student Award - HSC, awarded by KGISL for academic excellence
- HyperVerge Hackathon (Team of 2) - SensAI: Built learner engagement modules (streaks, quests, at-risk alerts)